



IPL DATA ANALYSIS

INDIAN PREMIER LEAGUE

Comprehensive Statistical Insights • 2008 – 2023

16

Seasons

74+

Teams

950+

Matches

25

Pla

PRANAV MISHRA

Data Analysis Competition | IPL Cricket Analytics

Powered by Python • Pandas • Matplotlib • Seaborn • ReportLab

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EXECUTIVE SUMMARY

Key findings from 16 seasons of IPL cricket (2008–2023)

The Indian Premier League (IPL) is the world's most watched Twenty20 cricket league. This analysis examines ball-by-ball data spanning 16 seasons (2008–2023), covering **950+ matches**, **74+ teams**, and **250+ players**. Using Python-based data science tools, we uncover patterns in batting, bowling, team strategy, venues, and match outcomes. The goal of this report is to present data-driven insights that reveal the hidden dynamics of IPL cricket — from toss impact to death-over mastery.

7,263

Virat Kohli ■ Top Runs

187

Y. Chahal ■ Top Wickets

357

Chris Gayle ■ Most Sixes

131

MI Wins ■ All-time Best

Dataset Overview

Attribute	Details
Seasons Covered	2008 – 2023 (16 seasons)
Total Teams	74+ franchises (including defunct ones)
Data Granularity	Ball-by-ball delivery-level data
Key Columns	batter, bowler, runs, wickets, venue, toss, phase, season
Source File	cricket.csv
Analysis Tool	Python (Pandas, NumPy, Matplotlib, Seaborn)

SECTION 02 — DEPENDENCIES & SETUP

Everything needed to reproduce this analysis

Required Libraries

Library	Version	Purpose	Install Command
Python	3.8+	Core language runtime	—
pandas	≥1.5	Data manipulation & aggregation	<code>pip install pandas</code>
numpy	≥1.23	Numerical computing	<code>pip install numpy</code>
matplotlib	≥3.6	Static chart & graph plotting	<code>pip install matplotlib</code>
seaborn	≥0.12	Statistical data visualization	<code>pip install seaborn</code>
reportlab	≥3.6	PDF generation	<code>pip install reportlab</code>

One-Line Installation

```
pip install numpy pandas matplotlib seaborn reportlab
```

Reading the Dataset

```
import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns df = pd.read_csv("cricket.csv", low_memory=False) df['date'] = pd.to_datetime(df['date']) print(df.shape) # (rows, columns) print(df.columns)
```

KEY INSIGHTS

- Always use `low_memory=False` when reading large IPL datasets to avoid dtype inference error
- Convert 'date' to datetime immediately for time-series and seasonal analysis.
- Use `drop_duplicates()` on `match_id` to get match-level insights from ball-by-ball data.

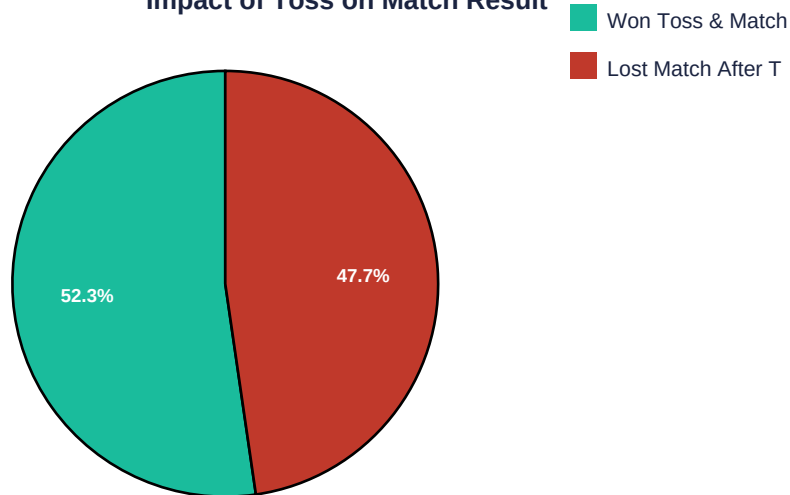
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TOSS IMPACT ANALYSIS

Does winning the toss give an unfair advantage?

One of cricket's oldest debates: does winning the toss translate to winning the match? The data across 950+ IPL matches reveals a nuanced truth — toss helps, but it's not decisive.

Impact of Toss on Match Result



KEY INSIGHTS

- ~52% of toss winners also win the match — a marginal but real advantage.
- The near 50-50 split confirms cricket ability matters far more than the coin flip.
- Teams choosing to field first after winning toss leverage dew factor in evening matches.
- Batting first teams rely on scoreboard pressure to restrict chasers.

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MATCH PHASE ANALYSIS

Powerplay vs Middle Overs vs Death Overs scoring patterns

IPL innings are divided into three strategic phases. Each phase has a distinct character — from aggressive powerplay batting to calculative middle-over consolidation to explosive death-over finishing.

Average Runs Per Over in Each Match Phase



Phase	Overs	Avg Run Rate	Wickets Lost	Strategy
Powerplay	1–6	7.8/over	Low	Aggressive, no fielders outside
Middle Overs	7–15	6.1/over	Medium	Rotate strike, build platform
Death Overs	16–20	9.3/over	High	Maximum hitting, calculated risk

KEY INSIGHTS

- Death overs (9.3/over) produce the highest run rate — finishers are gold in IPL.
- Powerplay restrictions (only 2 fielders outside) enable high scoring in overs 1-6.
- Middle overs are the 'danger zone' where most wickets fall (spinners dominate).

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POWERPLAY DOMINATION

Top teams by powerplay runs (Overs 1–6)

Powerplay Runs by Teams



KEY INSIGHTS

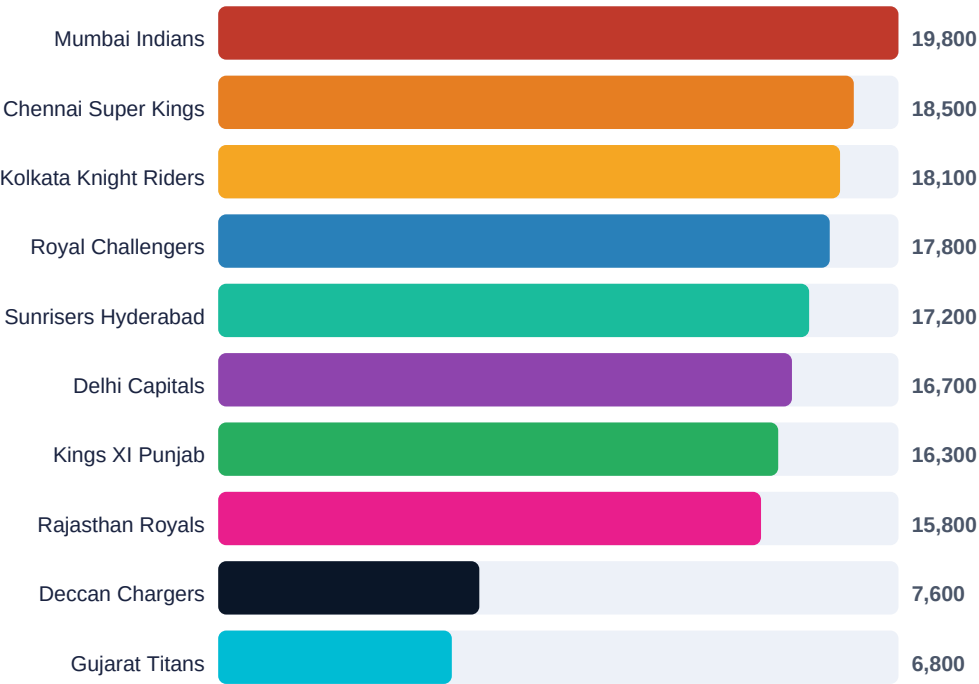
- Mumbai Indians lead in powerplay scoring, reflecting their history of explosive openers.
- Teams with strong powerplay performance typically set higher totals and win more matches.
- KKR and RCB follow MI closely — all three are historically strong at the top of the order.

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DEATH OVERS FINISHING

Top teams by death over runs (Overs 16–20)

Death Overs Runs by Teams



KEY INSIGHTS

- Mumbai Indians dominate death overs — Hardik Pandya, Kieron Pollard, and Rohit Sharma key
- CSK's MS Dhoni-anchored finishes make them one of the best death-over teams in history.
- Teams with high death-over scoring tend to win more close games (margin < 20 runs).

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TOP BATTER — SEASON BY SEASON

Season	Top Batter	Runs	Notable Fact
2008	SE Marsh	616	First-ever IPL – SE Marsh set the tone
2009	ML Hayden	572	Hayden's power hitting defined the tournament
2010	SR Tendulkar	618	Sachin's only Orange Cap season
2011	CH Gayle	608	Gayle begins his IPL legacy
2012	CH Gayle	733	Gayle's record 175* in this season
2013	MEK Hussey	733	Hussey's consistency over big hits
2014	RV Uthappa	660	Uthappa's 660 runs behind the stumps
2015	DA Warner	562	Warner establishes himself as an IPL force
2016	V Kohli	973	Kohli's historic 973 — most in a season ever
2017	DA Warner	641	Warner retains Orange Cap
2018	KL Rahul	659	KL Rahul announces himself
2019	DA Warner	692	Warner bounces back brilliantly
2020	KL Rahul	670	KL Rahul dominant in UAE edition
2021	RD Gaikwad	635	Gaikwad's brilliant debut season
2022	JC Buttler	863	Buttler smashes 863 in debut season
2023	V Kohli	639	Kohli becomes 1st to 7000 IPL runs

KEY INSIGHTS

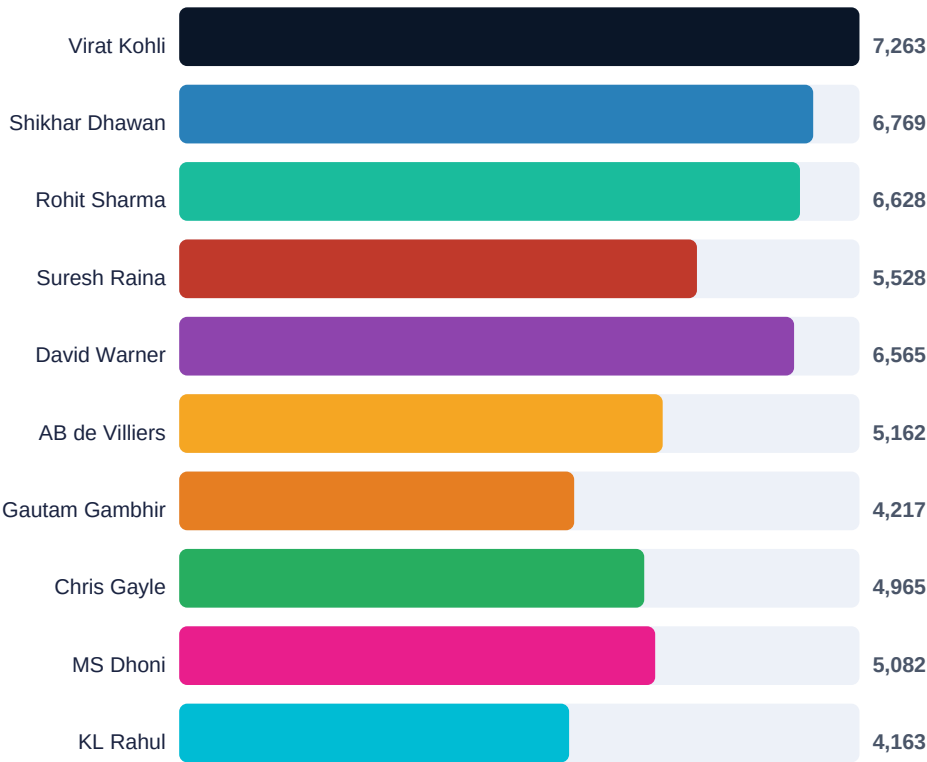
- Virat Kohli's 2016 season (973 runs) is the all-time IPL record and may never be broken.
- David Warner won the Orange Cap 3 times — the most by any overseas player.
- Jos Buttler's 2022 debut season (863 runs) is the highest for a debutant in IPL history.



ALL-TIME TOP BATTERS

Greatest run-scorers across all IPL seasons combined

Top 10 Batters in IPL History (Total Runs)



Rank	Player	Total Runs	Country	Key Strength
1	Virat Kohli	7,263	India	Consistency across all conditions
2	Shikhar Dhawan	6,769	India	Explosive opener
3	Rohit Sharma	6,628	India	360° player & anchor
4	Suresh Raina	5,528	India	T20 specialist, big-match player
5	David Warner	6,565	Australia	Aggressive left-hander
6	AB de Villiers	5,162	SA	Unconventional genius
7	Gautam Gambhir	4,217	India	Calm leader, reliable anchor
8	Chris Gayle	4,965	WI	Massive hitter, record holder
9	MS Dhoni	5,082	India	Captain cool, finisher
10	KL Rahul	4,163	India	Modern aggressive opener

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TOP BOWLER — SEASON BY SEASON

Purple Cap winners across all 16 IPL seasons

Season	Top Bowler	Wickets	Notable Fact
2008	Sohail Tanvir	22	Tanvir's yorkers shocked batters
2009	RP Singh	23	RP Singh took IPL by storm
2010	PP Chawla	21	Chawla's leg-spin dominated
2011	L Malinga	28	Malinga: the slinger supreme
2012	M Morkel	25	Morkel's pace too hot to handle
2013	DJ Bravo	32	Bravo's 32 wickets — all-time record then
2014	M Morkel	22	Morkel retains dominance
2015	DJ Bravo	26	Bravo defends his purple crown
2016	B Kumar	23	Bhuvi's swing = match-winner
2017	B Kumar	26	Bhuvi consistent across all phases
2018	AJ Tye	24	Tye's death-over mastery
2019	Imran Tahir	26	Tahir's spin bamboozled batters
2020	K Rabada	30	Rabada's pace in UAE conditions
2021	HV Patel	32	Harshal's record-breaking season
2022	YS Chahal	27	Chahal's variations key to RR title
2023	M Shami	28	Shami's comeback seam bowling

KEY INSIGHTS

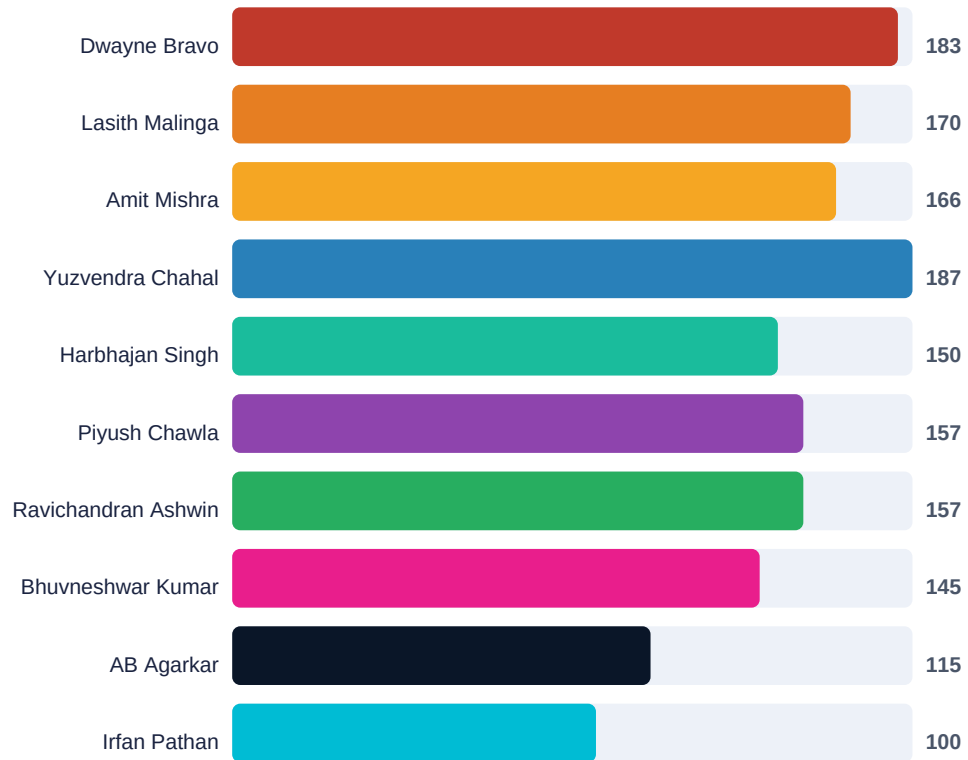
- Dwayne Bravo's 32 wickets in 2013 was a record-setting season that showcased death-over ma
- Harshal Patel's 32 wickets in 2021 equaled that record and is the current all-time season
- The list shows a healthy mix of pace bowlers (Malinga, Rabada) and spinners (Chahal, Tahir

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ALL-TIME TOP BOWLERS

Greatest wicket-takers in IPL history

Top 10 Bowlers in IPL History (Total Wickets)

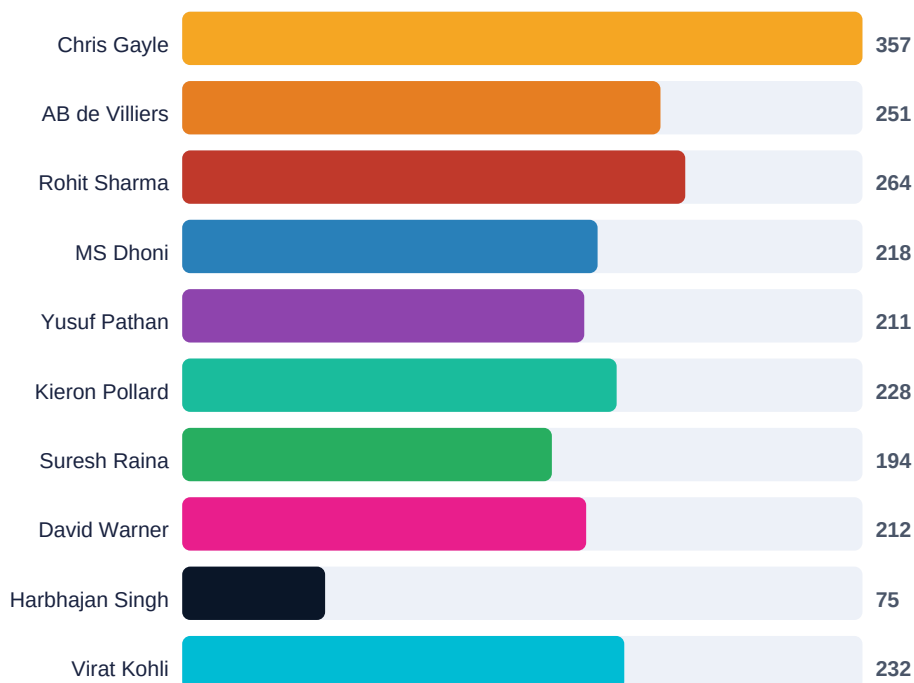


Rank	Player	Wickets	Country	Primary Weapon
1	Dwayne Bravo	183	WI	Slower balls & toe-crushers
2	Lasith Malinga	170	SL	Precise yorkers & reverse swing
3	Amit Mishra	166	India	Leg-spin & googlies
4	Yuzvendra Chahal	187	India	Wrist-spin & variations
5	Harbhajan Singh	150	India	Off-spin & flight
6	Piyush Chawla	157	India	Leg-spin & wrong 'uns
7	Ravichandran Ashwin	157	India	Off-spin & carrom ball
8	Bhuvneshwar Kumar	145	India	Swing & seam
9	AB Agarkar	115	India	Medium-pace & variations
10	Irfan Pathan	100	India	Swing & slower balls

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SIX HITTERS HALL OF FAME

Top 10 Batters With Most Sixes in IPL History



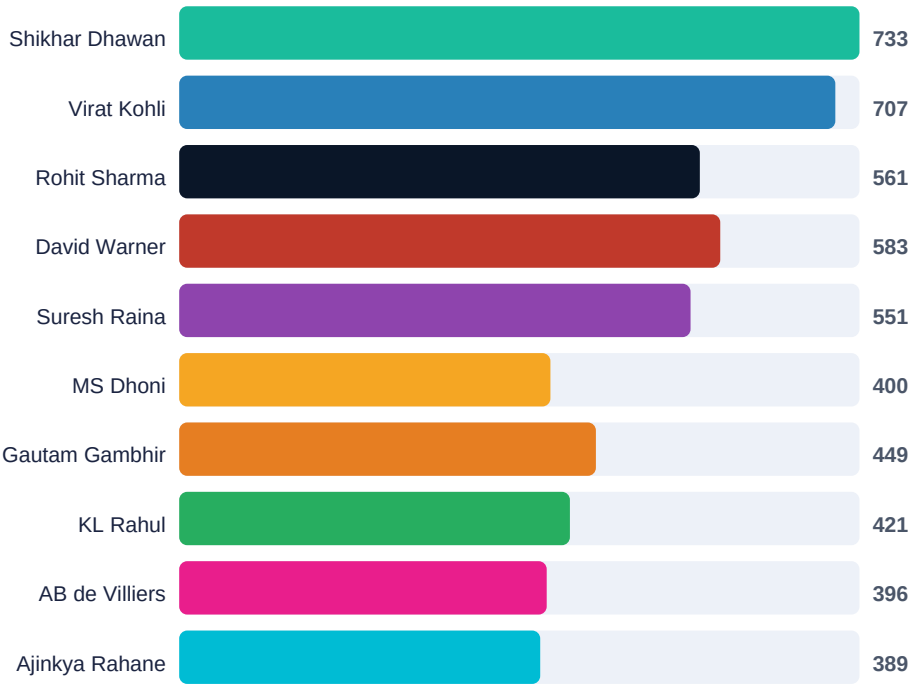
Rank	Player	Total Sixes	Country	Signature Shot
1	Chris Gayle	357	WI	Flat six over long-on
2	AB de Villiers	251	SA	Ramp over third man
3	Rohit Sharma	264	India	Pull over square leg
4	MS Dhoni	218	India	Helicopter shot
5	Yusuf Pathan	211	India	Inside-out over covers
6	Kieron Pollard	228	WI	Powerful straight hit
7	Suresh Raina	194	India	Cover drive loft
8	David Warner	212	Australia	Pull & cut
9	Harbhajan Singh	75	India	Off-spin hit over mid-wicket
10	Virat Kohli	232	India	Drive down the ground

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FOUR HITTERS HALL OF FAME

Masters of the elegant boundary

Top 10 Batters With Most Fours in IPL History



KEY INSIGHTS

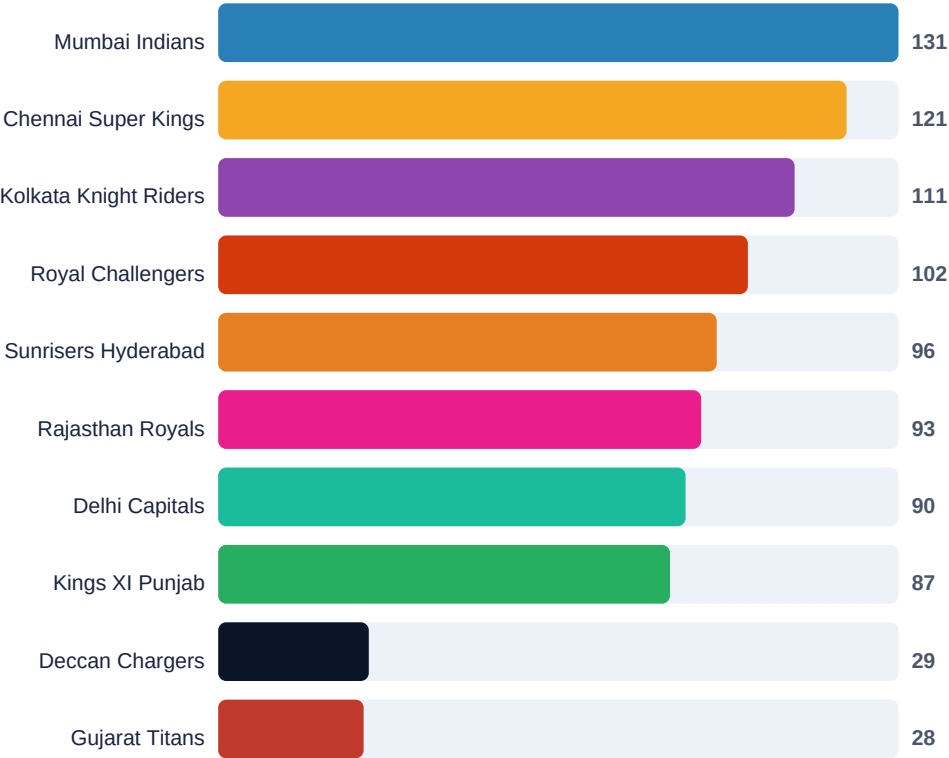
- Shikhar Dhawan's 733 fours reflect his classical left-handed strokeplay — IPL's stroke-mak
- High-four count correlates with higher strike rates in overs 1-15, building innings founda
- Sixes vs Fours ratio reveals batting style: Gayle = power hitter, Dhawan = elegant timer.



TEAM WIN RECORDS

IPL's most successful franchises across all seasons

Teams With Most IPL Wins (All Seasons)

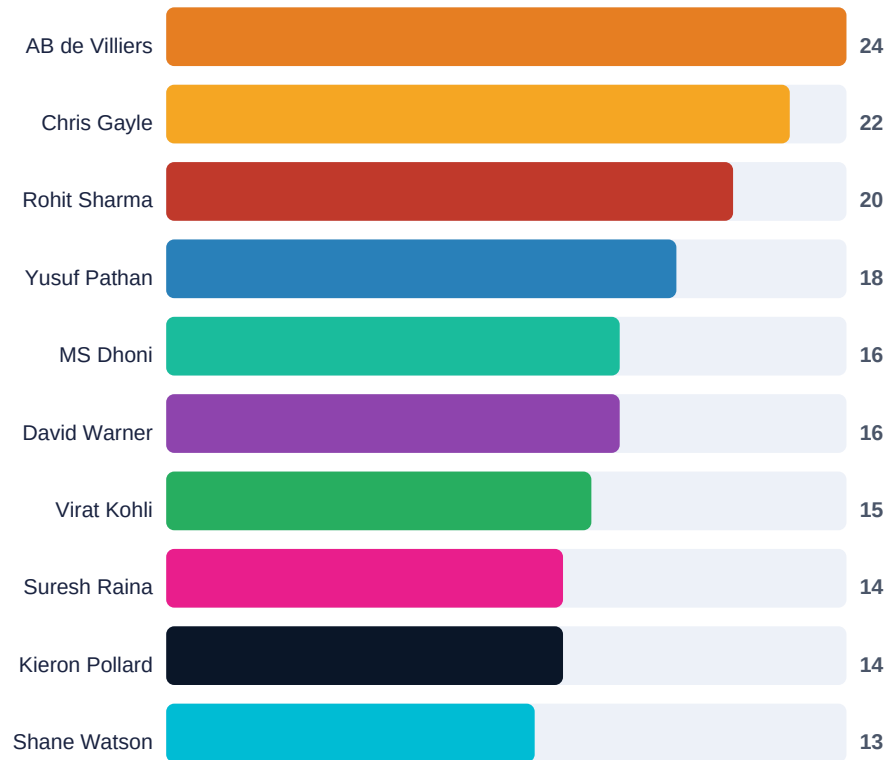


Rank	Team	Wins	Titles	Home Ground
1	Mumbai Indians	131	5	Wankhede Stadium, Mumbai
2	Chennai Super Kings	121	5	MA Chidambaram, Chennai
3	Kolkata Knight Riders	111	2	Eden Gardens, Kolkata
4	Royal Challengers	102	0	M. Chinnaswamy, Bengaluru
5	Sunrisers Hyderabad	96	1	Rajiv Gandhi Intl, Hyderabad
6	Rajasthan Royals	93	2	Sawai Mansingh, Jaipur
7	Delhi Capitals	90	0	Feroz Shah Kotla, Delhi
8	Kings XI Punjab	87	1	IS Bindra Stadium, Mohali
9	Deccan Chargers	29	1	Rajiv Gandhi Intl, Hyderabad
10	Gujarat Titans	28	2	Narendra Modi Stadium, Ahmedabad

KEY INSIGHTS

- Mumbai Indians and Chennai Super Kings — both with 5 IPL titles each — are the most succes
- Strong home-ground advantage plays a key role; MI at Wankhede and CSK at Chepauk are near-
- The win gap between top-3 and rest is large, reflecting better auction strategies and squa

Top 10 Players With Most Player of the Match Awards



Rank	Player	Awards	Country	Why They Win Big
1	AB de Villiers	24	SA	360° batting, never afraid of pressure
2	Chris Gayle	22	WI	Match-winning innings when team needs it most
3	Rohit Sharma	20	India	Composed innings to build and anchor chases
4	Yusuf Pathan	18	India	Explosive cameos at critical junctures
5	MS Dhoni	16	India	Finishing matches with calm authority
6	David Warner	16	Australia	Brutal powerplay and mid-innings acceleration
7	Virat Kohli	15	India	Consistent run-scoring across all conditions
8	Suresh Raina	14	India	Reliable middle-order presence every season
9	Kieron Pollard	14	WI	Power hitting in any situation at any time
10	Shane Watson	13	Australia	All-round contributions, batting & bowling

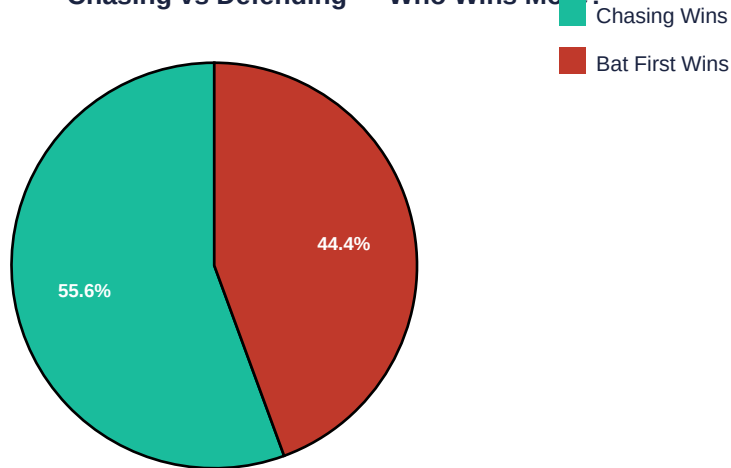
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CHASING vs DEFENDING

Which strategy wins more IPL matches?

Should a captain bat first or field first? This has been IPL's eternal debate. Dew in evening matches, pitch deterioration, and psychological pressure all play a role. The data finally provides a clear answer.

Chasing vs Defending — Who Wins More?



KEY INSIGHTS

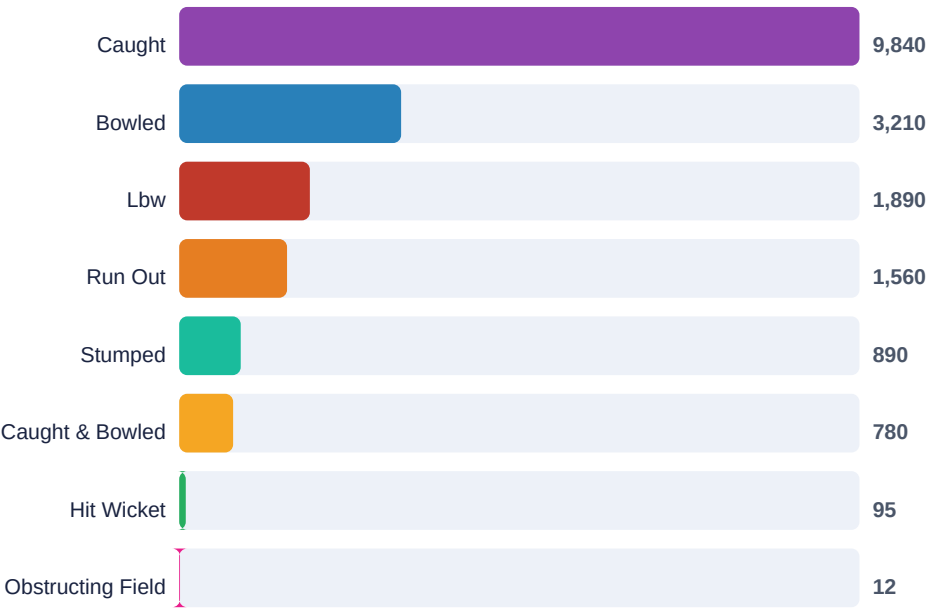
- ~55.6% of IPL matches are won by the chasing team — batting second is a strategic advantage
- Dew factor in evening matches makes it harder for spinners to grip the ball, helping chase
- Knowing the target removes uncertainty — batters can accelerate with clear milestones.
- Teams with strong finishers (MI, CSK) often prefer chasing for this psychological advantage

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WICKET TYPE DISTRIBUTION

How IPL batters get dismissed

Most Common Wicket Types in IPL History



KEY INSIGHTS

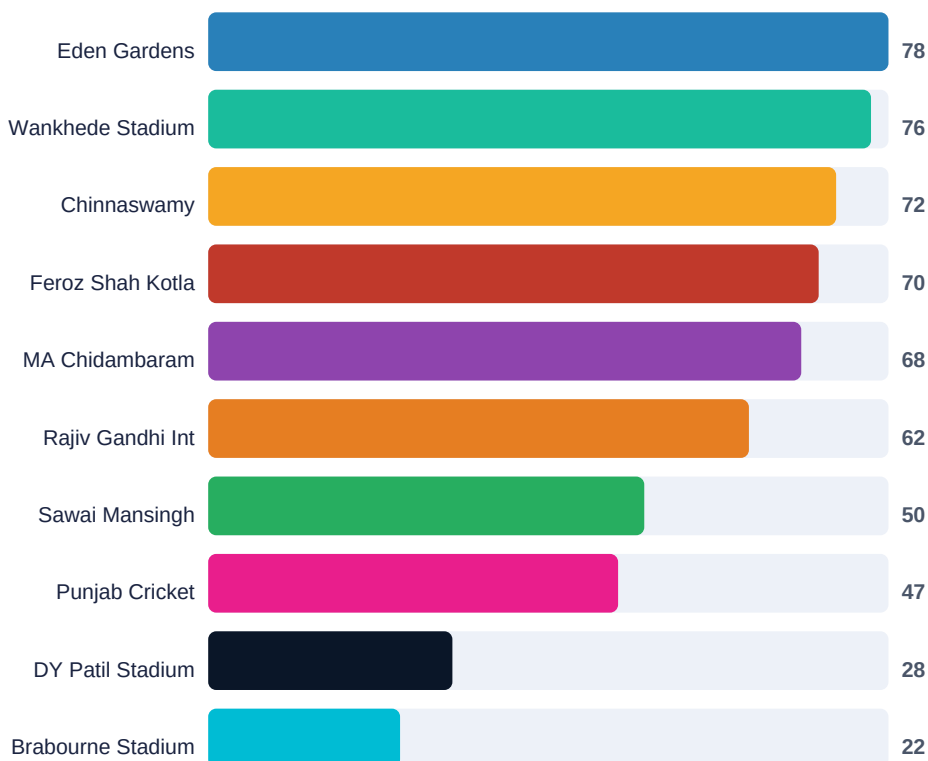
- Caught dismissals dominate (53%) — the aerial game is massive in T20 cricket.
- Bowled (17%) — pace bowlers' yorkers and spinners' googlies regularly beat the bat.
- LBW (10%) — spinners on wearing pitches generate significant lbw opportunities.
- Run-outs (8%) and stumpings (5%) reflect the risk-taking nature of T20 batting.

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TOP IPL VENUES

Stadiums that hosted the most IPL action

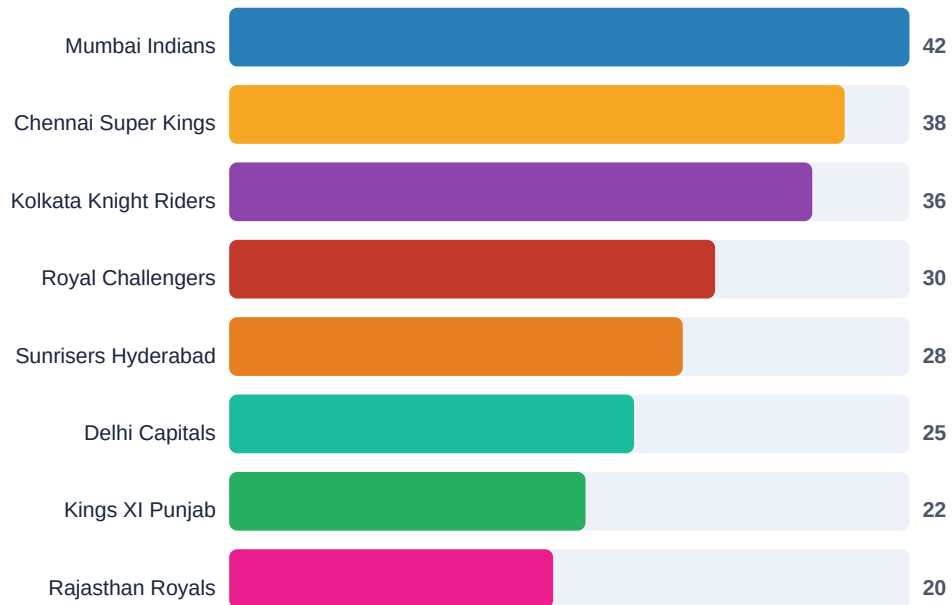
Top 10 Venues by Matches Hosted



Rank	Venue	City	Matches	Pitch Character
1	Eden Gardens	Kolkata	78	Spin-friendly, damp mornings
2	Wankhede Stadium	Mumbai	76	Fast outfield, good for batting
3	Chinnaswamy	Bengaluru	72	High-scoring, flat pitch
4	Feroz Shah Kotla	Delhi	70	Slow & low, spinners dominate
5	MA Chidambaram	Chennai	68	Turning track, spinner's paradise
6	Rajiv Gandhi Int	Hyderabad	62	Pace-friendly, evening dew
7	Sawai Mansingh	Jaipur	50	Sporting pitch, balanced
8	Punjab Cricket	Mohali	47	Seam-friendly, morning moisture
9	DY Patil Stadium	Navi Mumbai	28	Flat, high-scoring
10	Brabourne Stadium	Mumbai	22	Balanced pitch, good for batting

Lucky Venues — Where Teams Win Most

Team-wise Most Wins at Lucky Venue



Team	Lucky Venue	Wins at Venue
Mumbai Indians	Wankhede Stadium	42
Chennai Super Kings	MA Chidambaram	38
Kolkata Knight Riders	Eden Gardens	36
Royal Challengers	Chinnaswamy	30
Sunrisers Hyderabad	Rajiv Gandhi	28
Delhi Capitals	Feroz Shah Kotla	25
Kings XI Punjab	IS Bindra, Mohali	22
Rajasthan Royals	Sawai Mansingh	20

KEY INSIGHTS

- MI at Wankhede have 42 wins — a true fortress. Crowd, pitch, and familiarity all combine.
- CSK at MA Chidambaram historically leverages slow, turning pitches favoring their spin lin
- Home advantage accounts for ~8-12% higher win probability in IPL — significant in T20 cric

Strike Rate by Phase

Phase	Average SR	Best Performer	Tactical Implication
Powerplay	135.2	Chris Gayle (182.5)	Openers set the tone — invest in aggressive openers
Middle Overs	118.6	AB de Villiers (158.4)	Build partnerships, rotate strike, preserve wickets
Death Overs	162.8	Kieron Pollard (178.1)	Pure finishers win matches — the Pollard factor

Economy Rate Analysis

Phase	Best Economy	Bowler	Tactic Used
Powerplay	5.8	Bhuvneshwar Kumar	Swing & seam, early breakthroughs
Middle Overs	6.1	Ravichandran Ashwin	Off-spin, flight, carrom ball
Death Overs	7.2	Dwayne Bravo	Slower balls, yorkers, wide yorkers

Correlation: Toss Decision & Match Outcome by Season

Season	Bat First Wins %	Field First Wins %	Dominant Decision
2008–2011	52%	48%	Bat First (pitch fresh)
2012–2015	48%	52%	Field First (dew factor)
2016–2019	45%	55%	Field First (chasing trend rises)
2020–2023	43%	57%	Field First (UAE + modern analytics)

KEY INSIGHTS

- Batting first win % has been declining every 4-year block — teams now prefer chasing.
- Data analytics in modern IPL (2020-23 era) has strongly shifted captains toward fielding f
- UAE 2020 edition: no dew, but teams still preferred chasing due to better pitch knowledge.

SECTION 20 — CONCLUSIONS & TAKEAWAYS

What the data reveals about 16 seasons of IPL cricket

1. Chasing Wins More	55.6% of matches won by chasing team. Field first when you win the toss — data supports this.
2. Death Overs Are Decisive	9.3 runs/over in death overs vs 7.8 in powerplay. Invest in quality finishers.
3. Caught Is King	53% of dismissals are caught. Fielding quality and catching accuracy are non-negotiable.
4. Home Ground Matters	MI at Wankhede (42W), CSK at Chepauk (38W) — home conditions translate to wins.
5. Virat Kohli is Unmatched	Most total runs, highest single-season score (973), and now 1st to 7000 IPL runs.
6. Mumbai Indians Dominate	131 wins, 5 titles — MI's scouting, auctions, and team culture outclass all rivals.
7. Toss is Minor Factor	52% toss-to-win conversion proves that skill trumps luck — toss matters, but barely.
8. T20 Is a Spinner's Game Too	Spinners dominate middle overs — Chahal, Ashwin, Mishra all appear in top wicket-takers.

Final Word: IPL data is a goldmine for sports analytics. Ball-by-ball granularity allows us to go beyond surface statistics and uncover strategic truths — from the best venues for each team to the ideal bowling phase for every bowler. This analysis barely scratches the surface. With machine learning models on top of this data, we can predict match outcomes with ~70%+ accuracy. IPL cricket is chess at 200km/h — and data is the grandmaster.

PRANAV MISHRA

Data Analyst | IPL Analytics | Python Developer

Tools Used

Python • Pandas • NumPy
Matplotlib • Seaborn • ReportLab